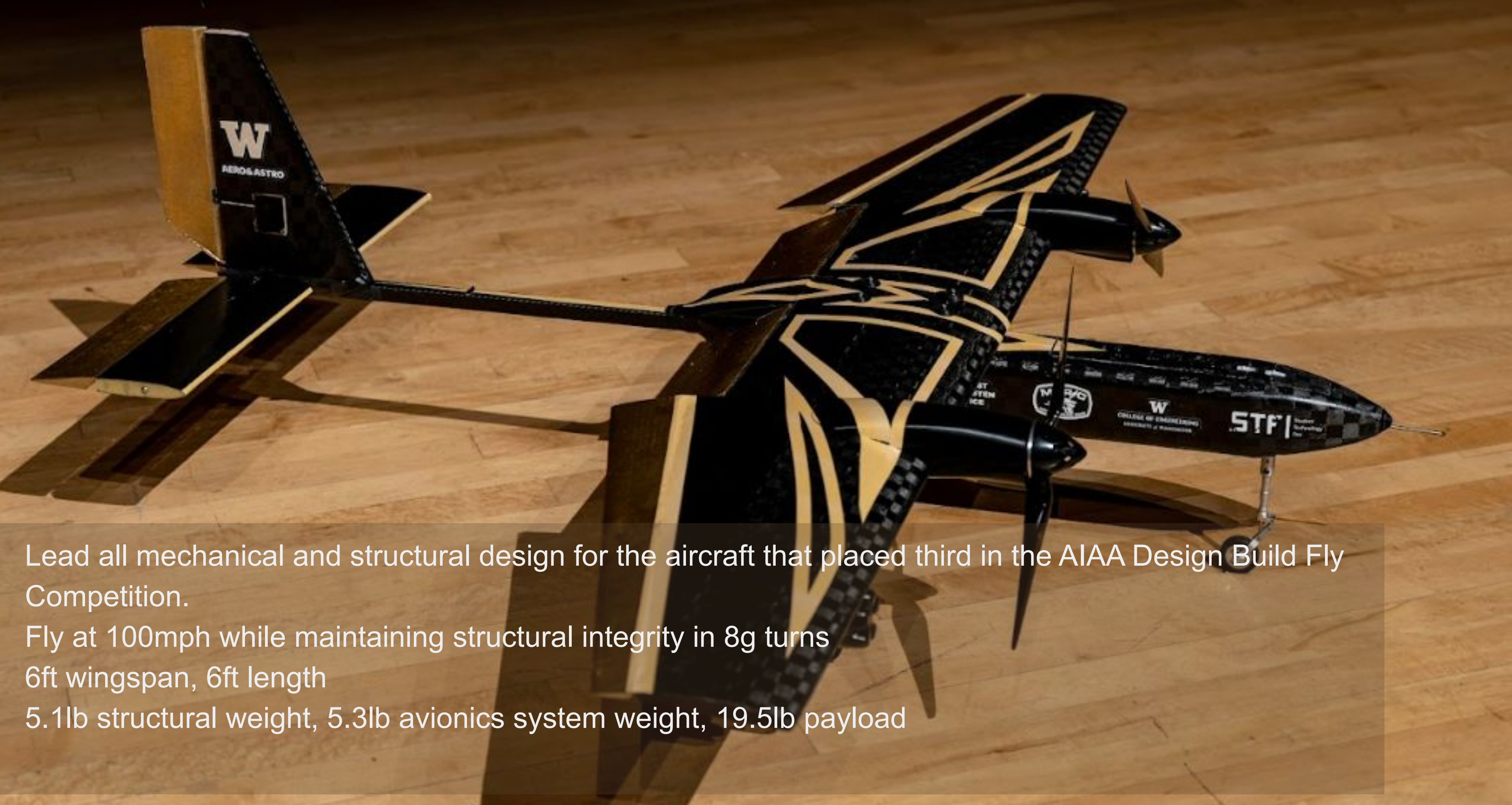
AIAA Design Build Fly at UW

Structures Lead 2022-2025 * Fuselage Project Lead 2022-2023



Lead all mechanical and structural design for the aircraft that placed third in the AIAA Design Build Fly Competition.

Fly at 100mph while maintaining structural integrity in 8g turns

6ft wingspan, 6ft length

5.1lb structural weight, 5.3lb avionics system weight, 19.5lb payload

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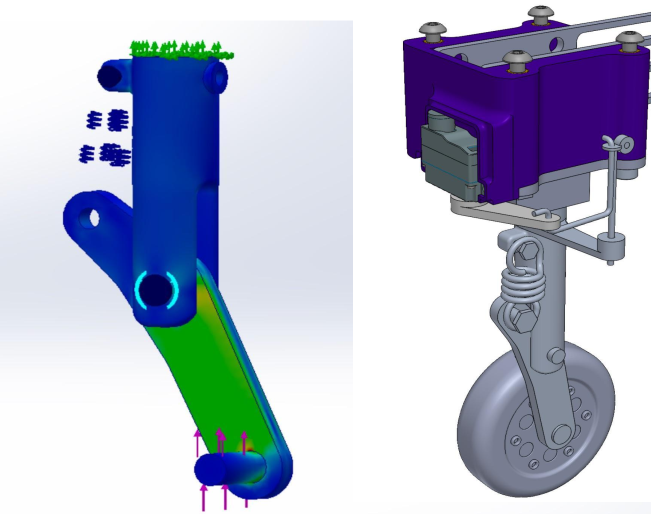
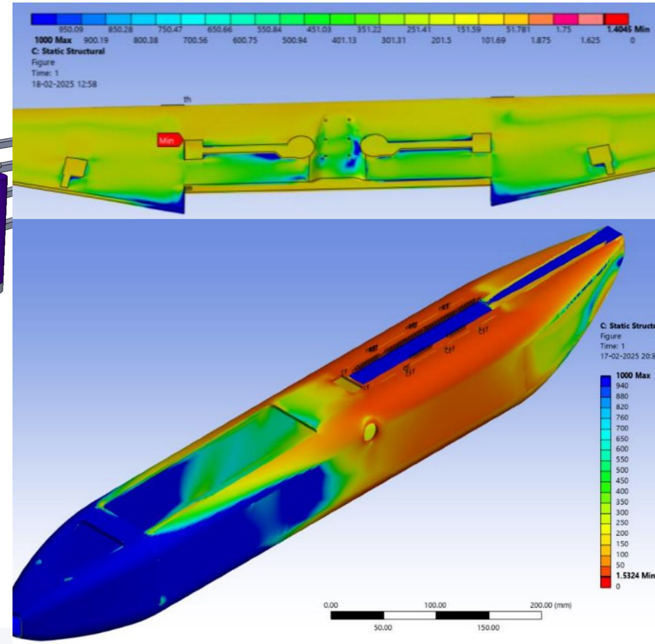
Selected Projects in DBF





Custom Retractable Landing Gear

- **Goal**: Design landing gear while surviving 6g landings
- **Result**: Utilizes a COTS retracting linear actuator with custom strut
- **Result**: Features steerable system

Composite Sandwich Panel FEA

- **Goal**: Optimize fuselage and wing to be within Factor of Safety (FOS)
- **Result**: Reduction in fuselage weight by 53% (450g->210g)
- **Result**: Optimized carbon fiber ply layers to maintain integrity

Development Projects

- **Goal**: Research potential new technology
- **Result**: Designed fowler flap with 25% chord extension
- **Result**: Designed 33% lighter landing gear (250g->165g)

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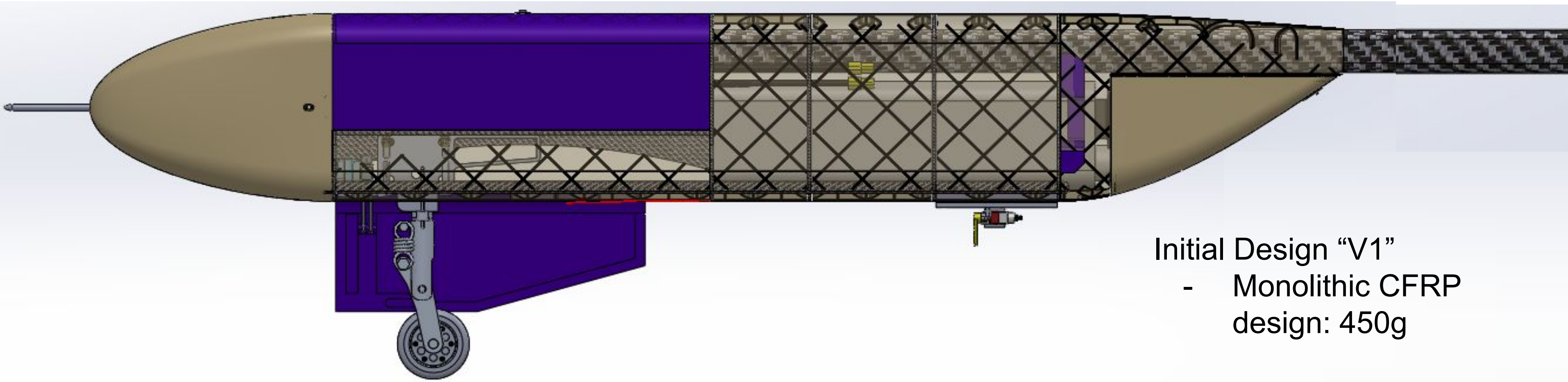
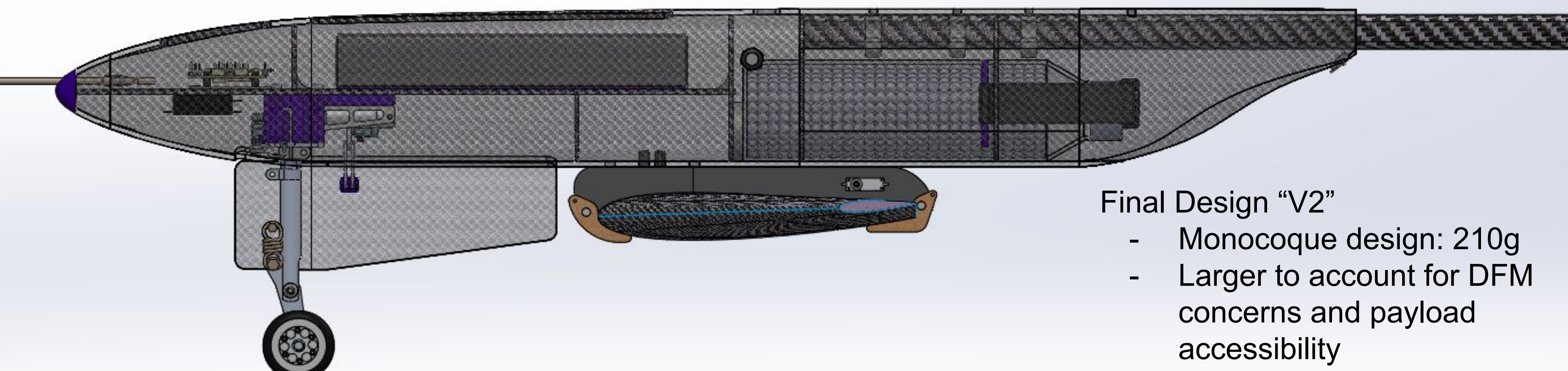
Development Cycle - Fuselage



Initial Design “V1”
- Monolithic CFRP design: 450g



Final Design “V2”
- Monocoque design: 210g
- Larger to account for DFM concerns and payload accessibility

Reduced the fuselage weight by over 50% by using CFRP Nomex panels and a multitude of hand calculations.

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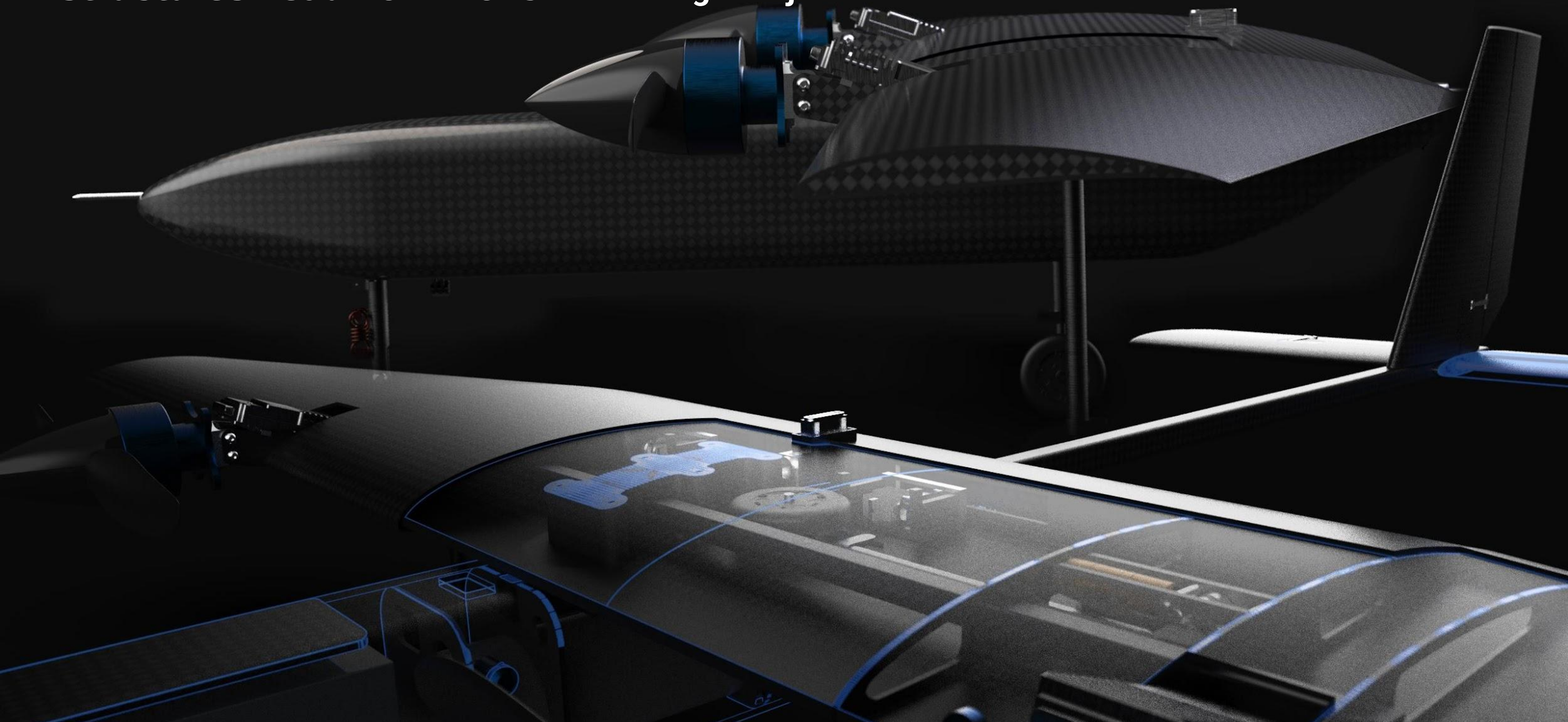
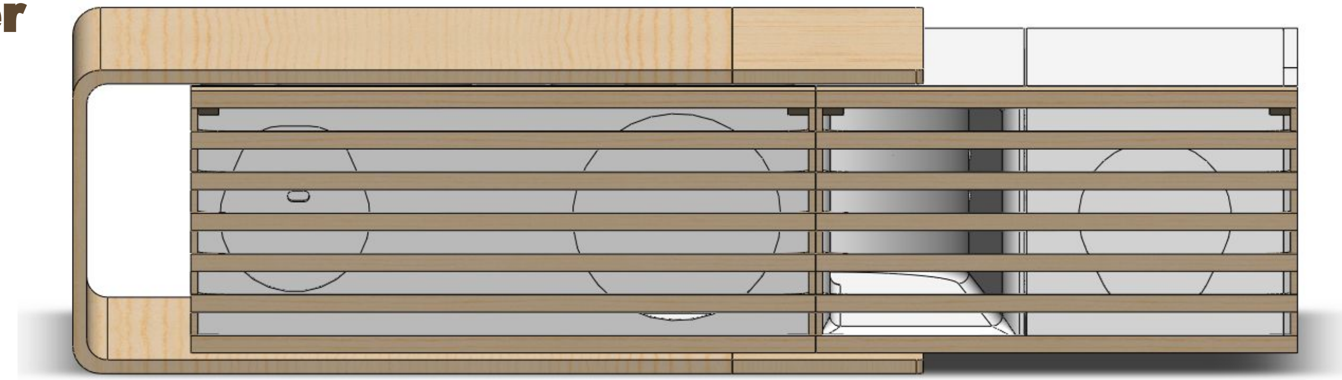
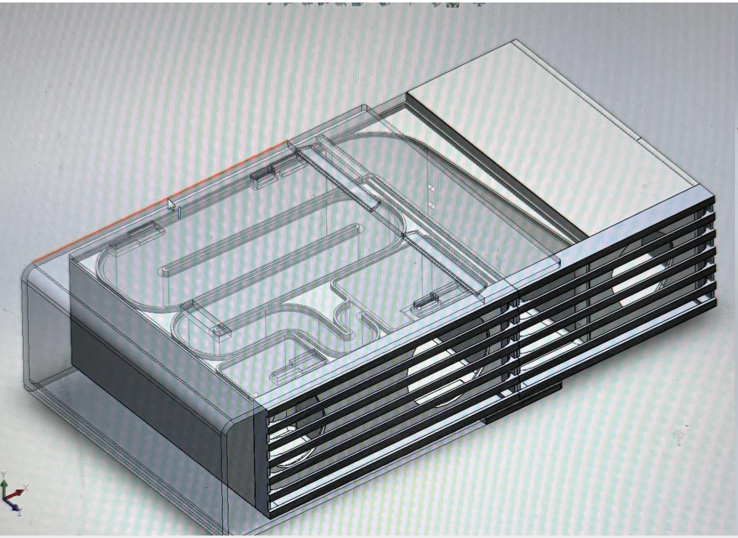


- Wings needed to integrate landing gear assembly and propulsion systems within the 1in airfoil thickness
- Milled aluminum parts on mounts and wing to fuselage joint

3D-Printed Speaker

Personal Project 2023

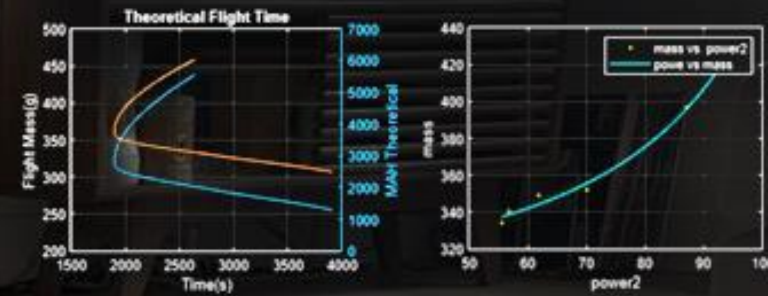




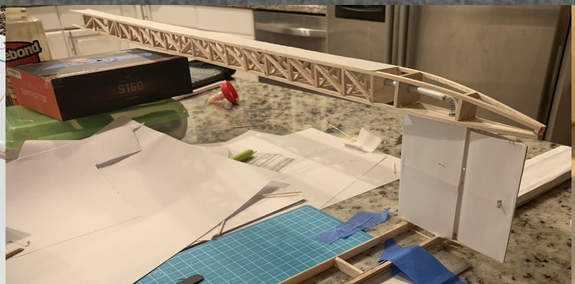
- Developed Plastic 3d printed wood that looks realistic and is easy to manufacture
- Preformed DFM principles to achieve assembly time of ~30 minutes
- Designed sound profile using acoustical horn analysis

Solar Aircraft

Personal Project 2020-2022



- Goal to as long as possible
- Used propulsion and flight data to make informed design decisions based of engineering principles
- Achieved over an hour flight time by iterating, learning and optimizing multiple versions.

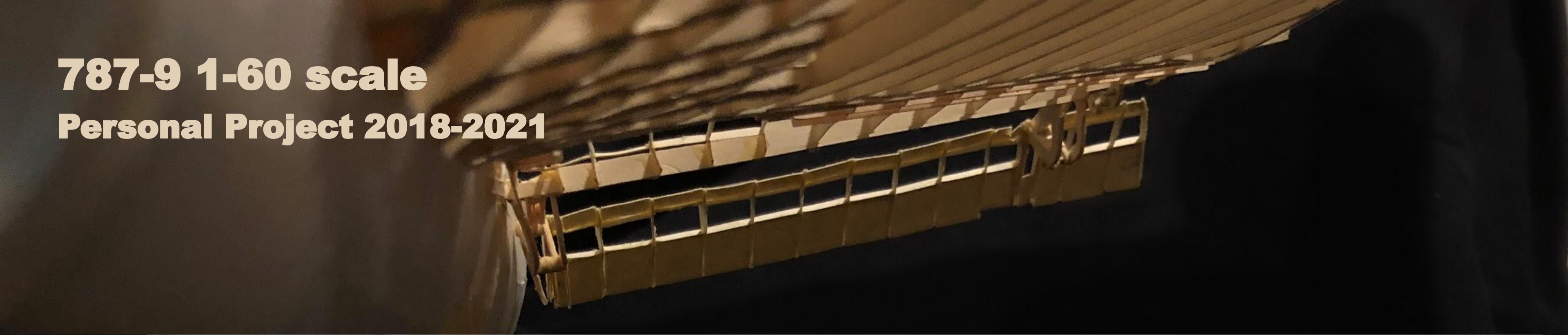


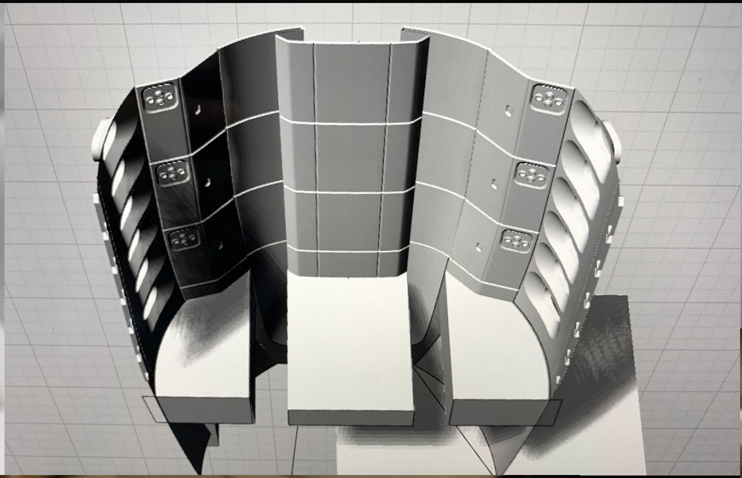
Span 2 Stress Test

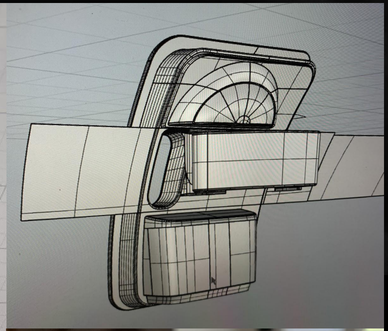
-190% stress at 120g, 40 mins

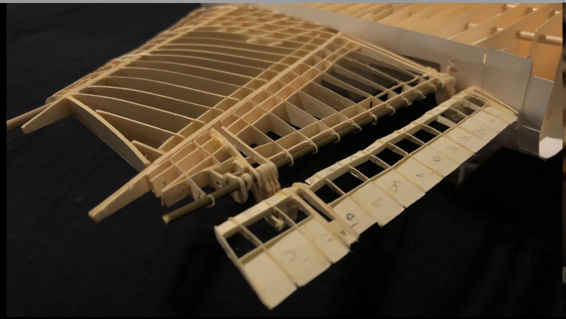
787-9 1-60 scale

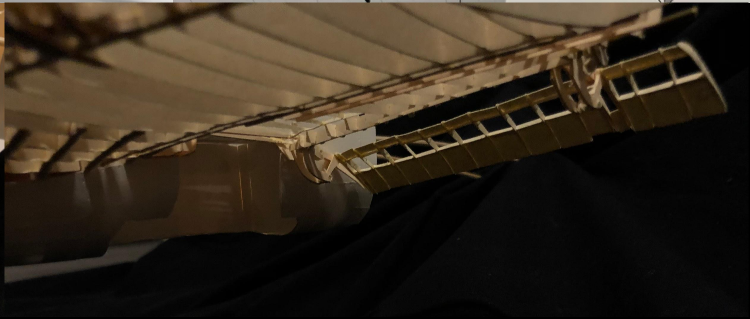
Personal Project 2018-2021











THANK YOU

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